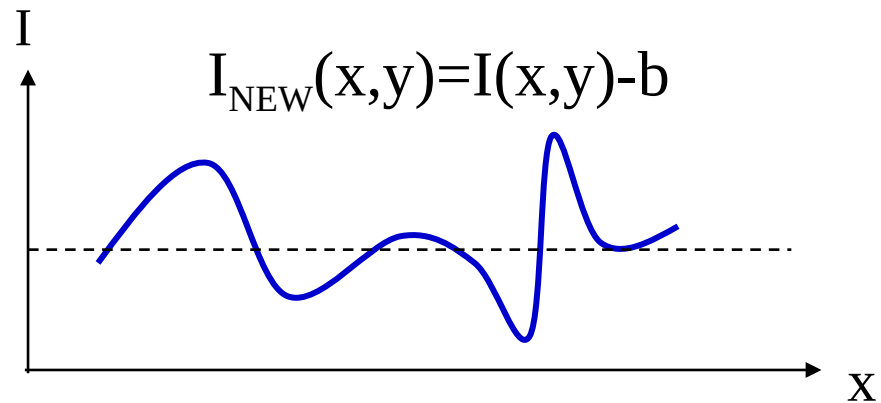
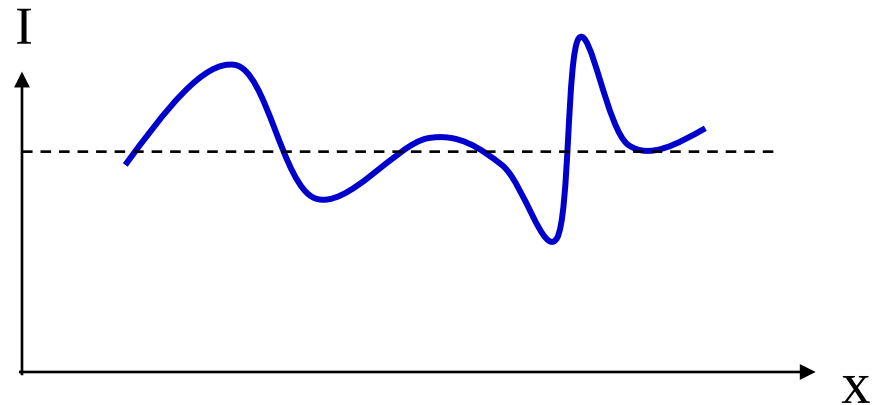


The Image Histogram



Image Mean

$$I_{av} = \frac{\sum_i \sum_j I(i, j)}{\sum_i \sum_j 1}$$



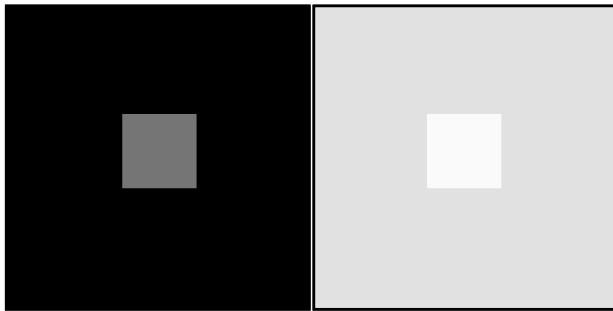


Changing the image mean

Image Contrast

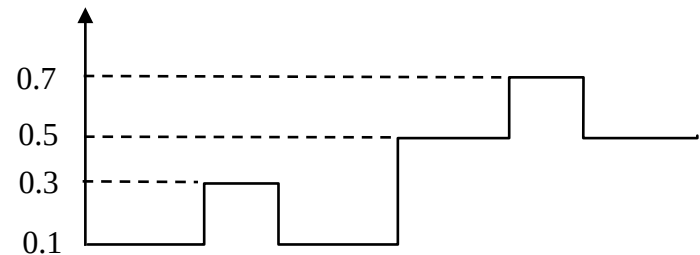
- The **local contrast** at an image point denotes the (relative) difference between the intensity of the point and the intensity of its neighborhood:

$$C = \left| \frac{I_p - I_n}{I_n} \right|$$



$$C = \left| \frac{0.3 - 0.1}{0.1} \right| = 2$$

$$C = \left| \frac{0.7 - 0.5}{0.5} \right| = 0.4$$



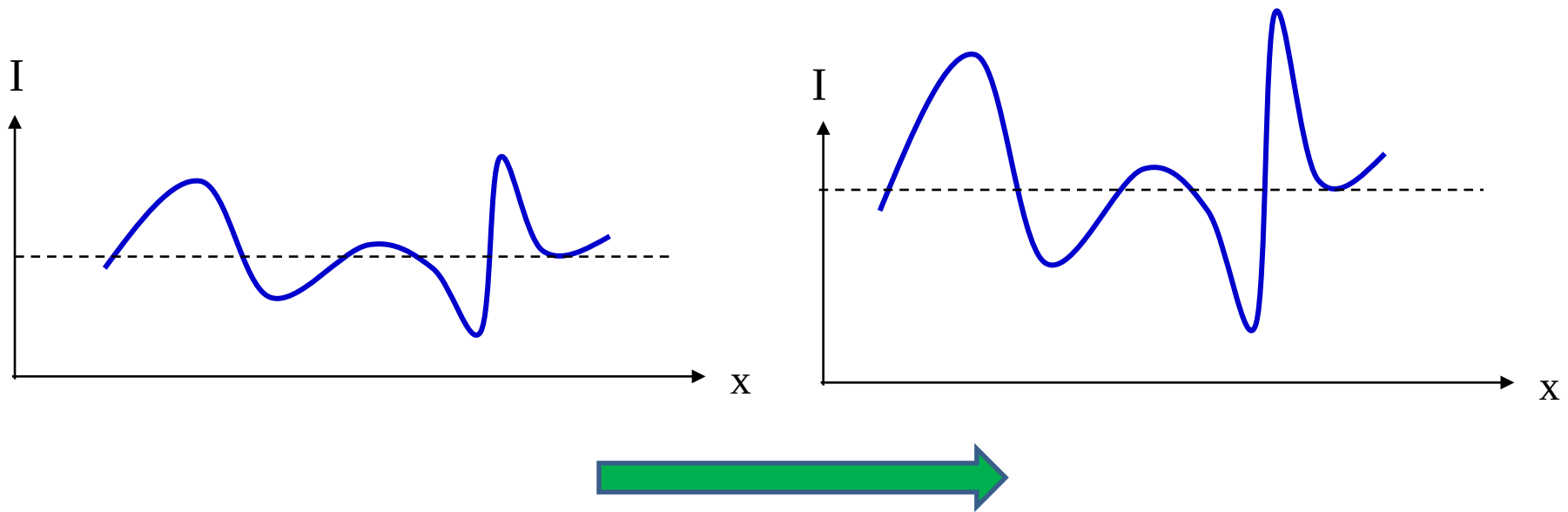
- The contrast definition of the entire image is ambiguous
- In general it is said that the image contrast is high if the image gray-levels fill the entire range



Low contrast



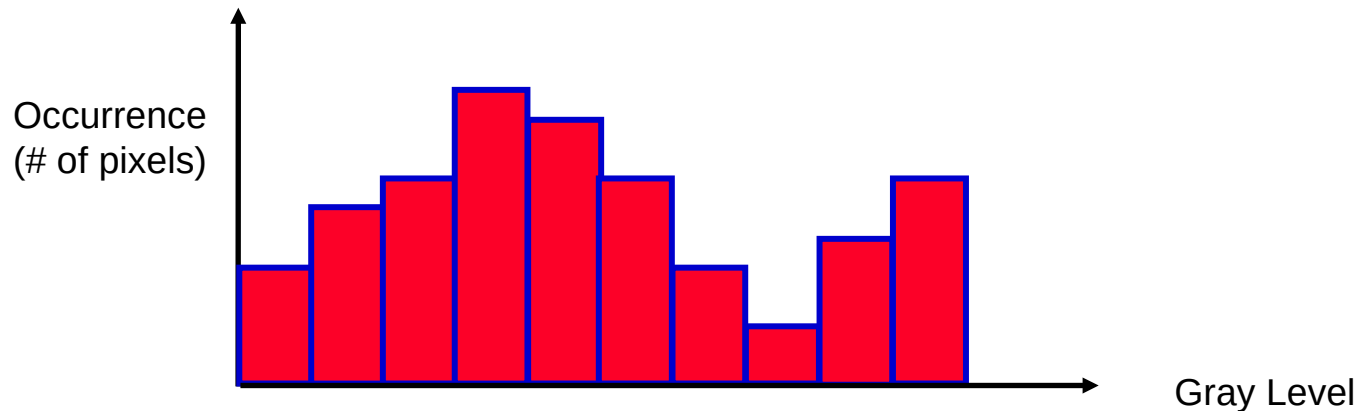
High contrast



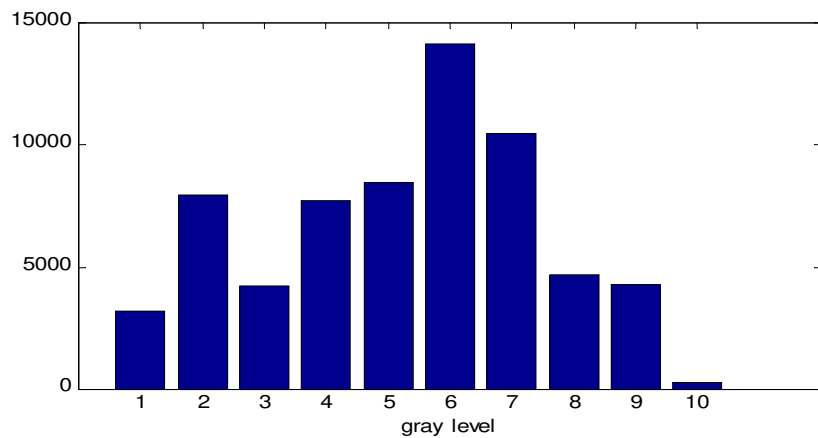
$$I_{\text{NEW}}(x,y) = \alpha \cdot I(x,y) + \beta$$

- How can we maximize the image contrast using the above operation?
- Problems:
 - Global (non-adaptive) operation.
 - Outlier sensitive.

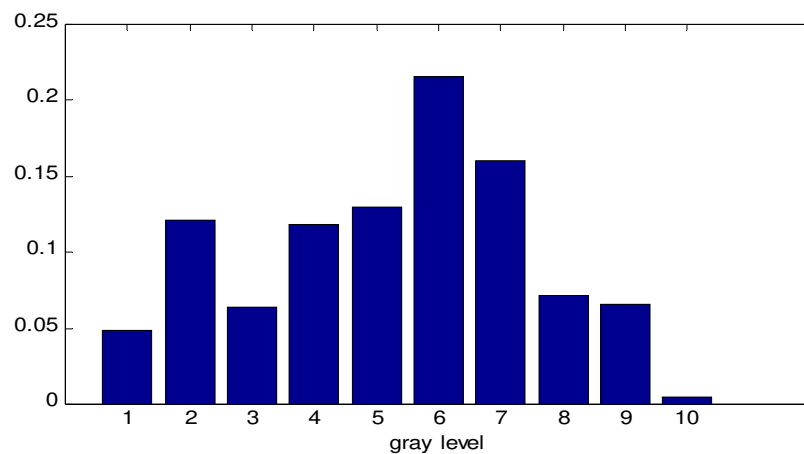
The Image Histogram



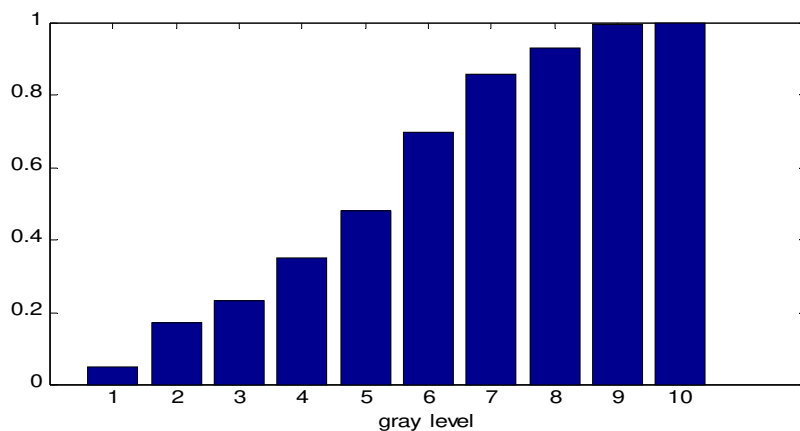
- $H(k)$ specifies the # of pixels with gray-value k
- Let N be the number of pixels: $N = \sum_k H(k)$
- $P(k) = H(k)/N$ defines the normalized histogram
- $C(k) = \sum_{i=1}^k H(i)$ defines the accumulated histogram



Histogram

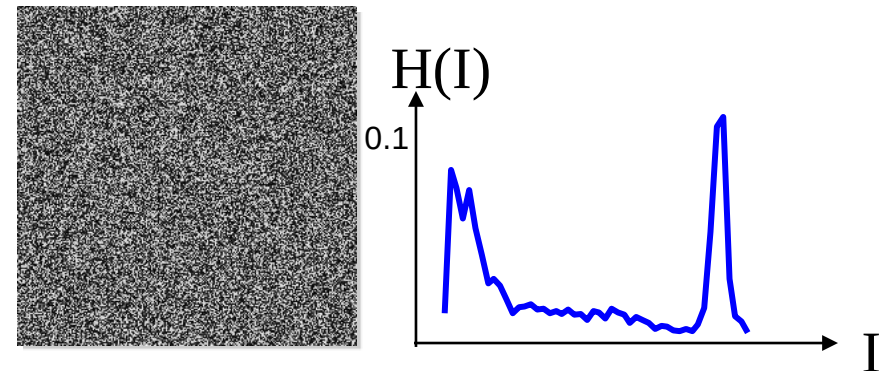
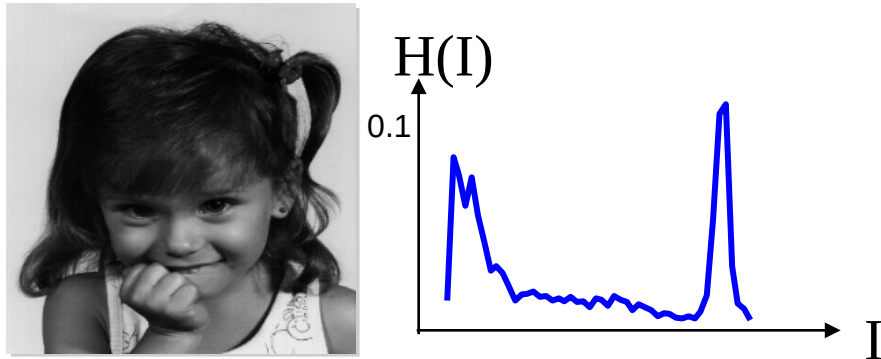
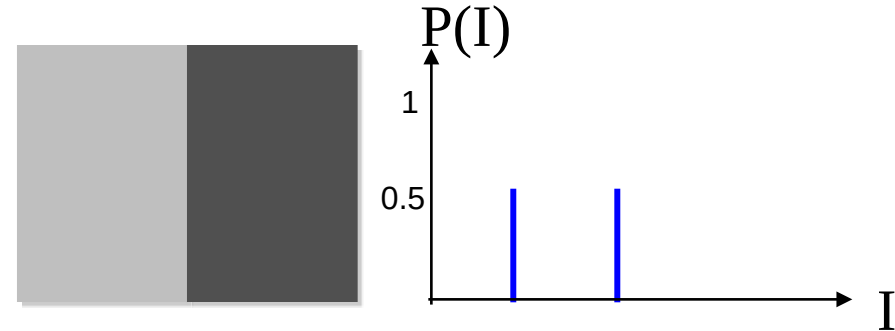
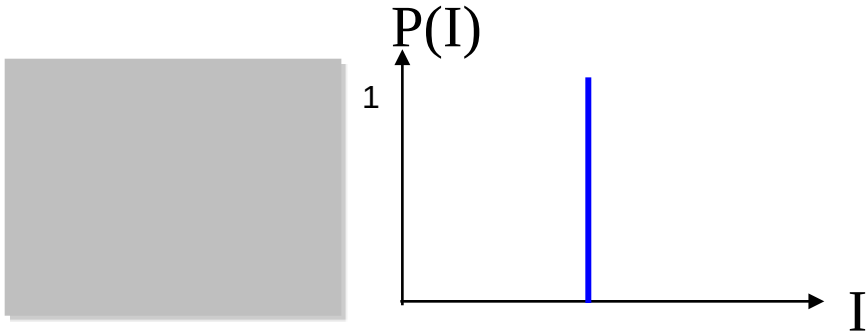


Normalized Histogram



Accumulated Histogram

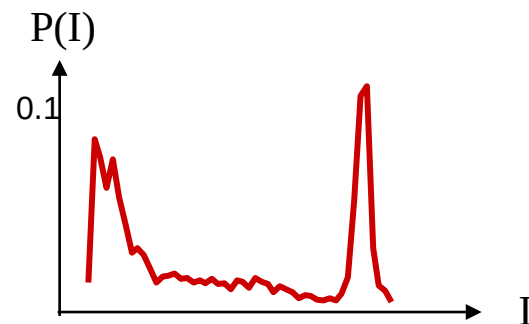
Examples



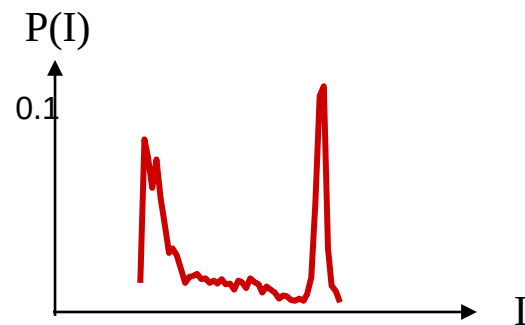
Pixel permutation of
the left image

- The image histogram does not fully represent the image

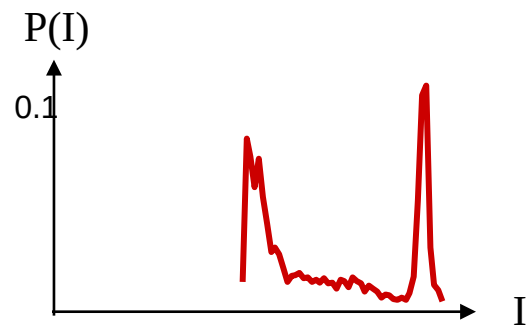
Original image



Decreasing contrast



Increasing average

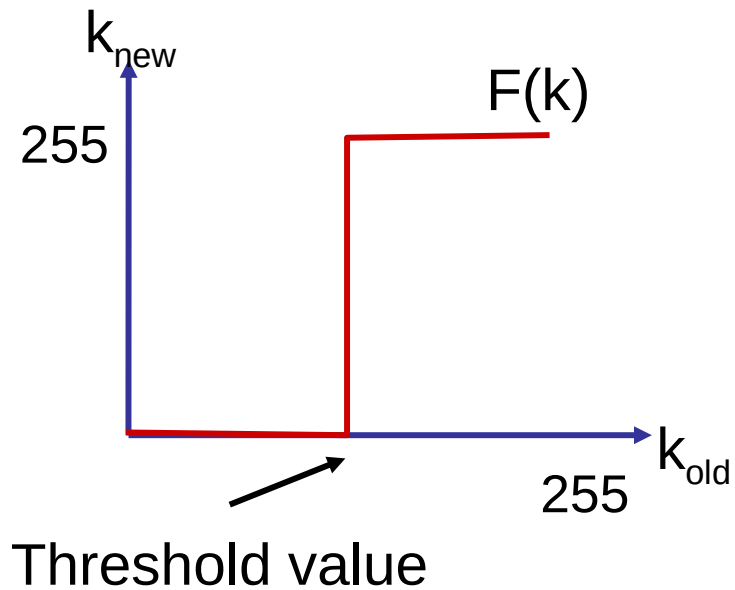


Adaptive Histogram

- In many cases histograms are needed for local areas in an image
- Examples:
 - Pattern detection
 - adaptive enhancement
 - adaptive thresholding
 - tracking



Thresholding

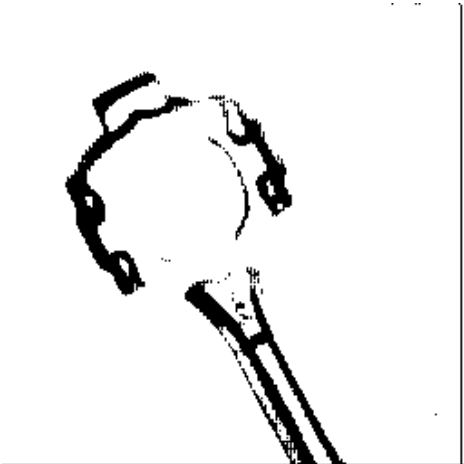


Threshold Selection

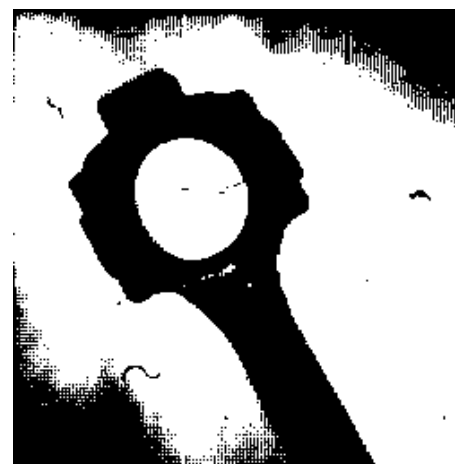
Original Image



Binary Image



Threshold too low



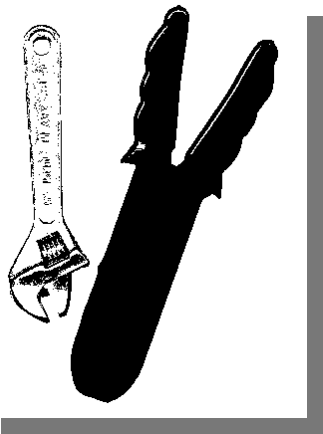
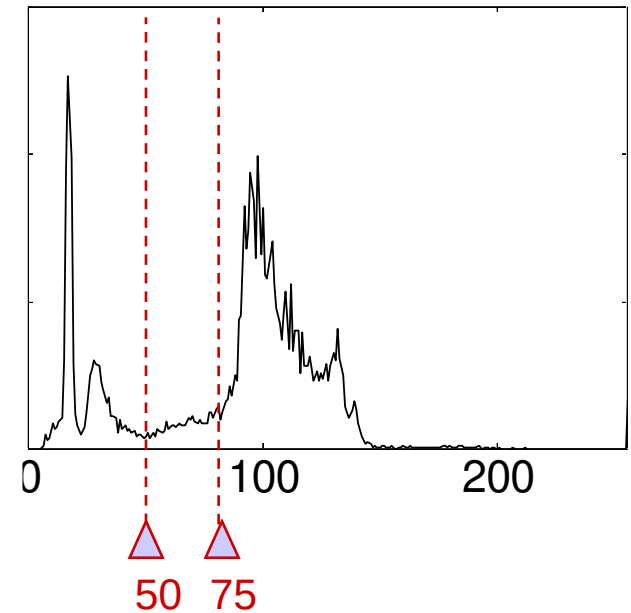
Threshold too high

Segmentation using Thresholding

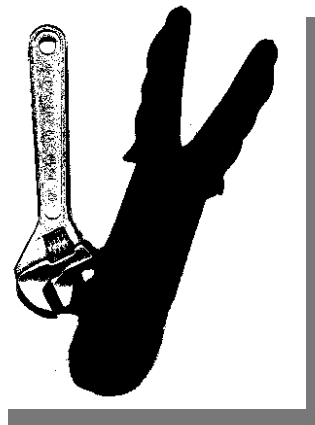
Original



Histogram



Threshold = 50



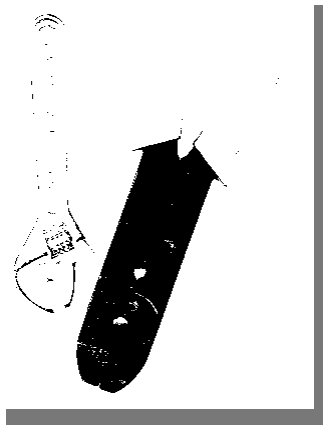
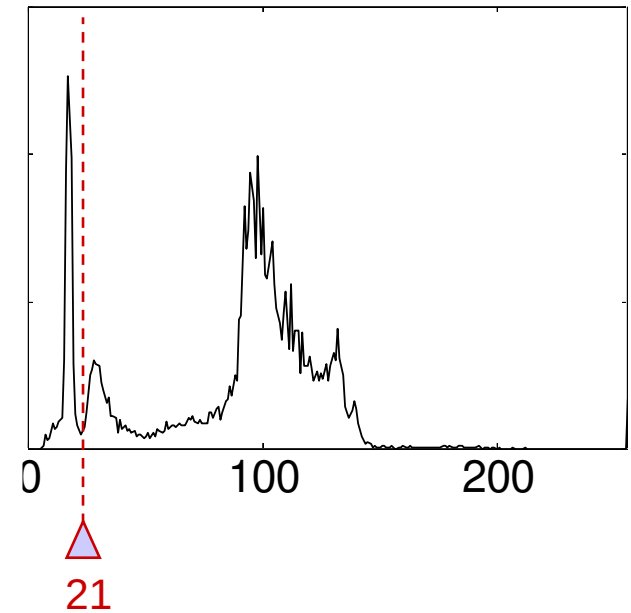
Threshold = 75

Segmentation using Thresholding

Original



Histogram



Threshold = 21

Adaptive Thresholding

- Thresholding is space variant.
- How can we choose the the local threshold values?

